

AKSHAY ASHOK

M.Sc. Data Science & Artificial Intelligence | Saarland University

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PROFILE

Master's student in Data Science & AI (4th semester) who ships measured, reproducible ML: a first-author Springer publication in computer vision, an Arabic-to-German OCR/translation benchmark study for a Gov-Tech client and a deployed NLP scoring service (FastAPI, Docker, CI). Experienced across the full pipeline, from dataset construction and model benchmarking to evaluation design, with LLM APIs and agentic tooling in daily use. Open to a Master's thesis topic from late 2027 and research collaborations.

EXPERIENCE

Student Consultant (AI / Data Science) · Lingomatch GmbH · Saarbrücken

Apr 2026 – Jul 2026

- Built and benchmarked the ML backend of a document-triage feasibility study for a Gov/Justice-Tech language-services platform (UdS Innovation Center, Triathlon case-study programme): turning photos of Arabic documents into usable German gist translations within an 8 GB laptop-GPU budget.
- Evaluated Surya OCR on KITAB-Bench (per-sample CER/WER, confidence calibration); tested PaddleOCR as an independent second engine for disagreement-based flagging and documented why it fails on Arabic.
- Benchmarked Opus-MT vs. NLLB vs. TranslateGemma for translation; caught the licence issue (CC-BY-NC) ruling the best baseline out of production and recommended an Apache-2.0 model with comparable BLEU at 1/9 the memory.
- Designed confidence-based failure routing that flagged 4/4 catastrophic OCR failures, defining the pipeline's human-escalation path; presented the live demo in the final case-study presentation to the company's CEO.

github.com/Axwolf13/arabic-doc-triage-research · write-up: axwolf13.github.io/writing/arabic-ocr-benchmarks

Research Intern (Computer Vision) · National Institute of Technology, Puducherry

Jan – Apr 2024

- Built a real-time yoga pose estimation pipeline with a hybrid MoveNet Thunder (training) + MoveNet Lightning (inference) architecture: 11 asanas at 98.8% test accuracy (98.4% verified on the shipped quantized model) at 30+ FPS on consumer hardware; curated the 3,435-image dataset end to end.
- First-authored a peer-reviewed publication in *Sādhanā* (Indian Academy of Sciences, Springer, 2025); released the code, the trained model and a live in-browser demo (TensorFlow.js) with a one-command verification script.

doi.org/10.1007/s12046-025-02788-w · live demo: axwolf13.github.io/demo

Web Development Intern · Zoho Corporation, Chennai, India

Aug – Sep 2023

- Built a full-stack library management web app with a content-based book recommender (HTML/CSS, JavaScript, SQL).

SELECTED PROJECTS

MolCA reproduction: which input does a molecule LLM believe? · graduate seminar, Prof. Klakow

- Reproduced MolCA (EMNLP 2023) on two machines: 62.32 and 62.77 BLEU-2 against the paper's 62.0, 93.8% of captions character-identical, all 32 published retrieval metrics within 0.5 points; 14 patches to the code, several for unreported bugs.
- Designed the modality-conflict test the paper never ran: given one molecule's SMILES and another's graph, the model follows the graph about 90% of the time; a wrong graph costs 2.5× what wrong text does (paired bootstrap, every interval excludes zero); controlled for 49% train-test contamination on a second dataset.

github.com/Axwolf13/molca-reproduction · write-up: axwolf13.github.io/writing/molca

DODI: deception scoring for Terms-of-Service documents, shipped as a live web service

- Deterministic index of how hard a ToS hides that "Buy now" means "revocable licence"; temporal study of 10 platforms across 2015–2024, compared against ToS;DR expert grades.
- Cross-checked with two LLM judges (rubric-based structured scoring, self-consistency repeats): they barely agree with the index ($\rho = 0.12$, $n = 51$), exposing a false-positive mode. Audited my own study, fixed a counting bug and a mis-specified validation run, published a corrected v1.1.
- Deployed as a Dockerized FastAPI service with tests and CI (dodi-web.onrender.com) plus an MCP server over Streamable HTTP, tested as a tool in a Microsoft Copilot Studio agent.

Shadow Correction: piracy as a market-correction signal · Data and Society, UdS · team of 3

- Measured streaming-market fragmentation (Herfindahl–Hirschman Index); piracy-intent indices across 90+ countries isolate fragmentation, not affordability, as the driver.

EDUCATION

M.Sc. Data Science & Artificial Intelligence · Saarland University

Oct 2024 – Present

Coursework: Statistics with R, Human-Computer Interaction, Data and Society. Seminar: ML for Language Processing (MolCA).

B.Tech. Artificial Intelligence & Data Science · Bannari Amman Institute of Technology

Oct 2020 – Apr 2024

CGPA: 8.65/10. Thesis: LipNet-based visual speech recognition as a personal communication aid for hearing-impaired users.

TECHNICAL SKILLS & LANGUAGES

Languages & tools: Python, R, SQL, JavaScript; PyTorch, TensorFlow/Keras, Hugging Face Transformers, scikit-learn, spaCy, OpenCV, pandas, NumPy; FastAPI, Docker, CI/CD (GitHub Actions), Git, LaTeX.

LLM work: OpenAI, Anthropic & Gemini APIs, prompt design, structured outputs, LLM-as-judge evaluation, MCP server development, embedding-based retrieval, agentic workflows (Claude Code, Cursor).

Methods: model benchmarking and evaluation design (BLEU, CER/WER, confidence calibration), classical ML & deep learning, mixed-effects models, hypothesis testing, time-series and event-study design, NLP and OCR pipelines, web scraping.

Spoken languages: English (C1), German (A2, targeting B1), Tamil (native), Hindi (basic).